❤️ 10-Year Heart Disease Risk Predictor

Enter the patient's information below to estimate their risk of developing coronary heart disease (CHD) within the next 10 years, based on the Framingham Heart Study dataset.

Demographics | Medical History | Vitals & Labs

Sex

○ Female ◉ Male

Age

50 − +

Education Level (1 = lowest, 4 = highest)

3 − +

Predict CHD Risk

✅ **Low Risk** of CHD within 10 years. (Estimated probability: 31.0%)

This is a statistical estimate based on the Framingham dataset and is not a medical diagnosis. Please consult a healthcare professional.

ℹ️ About this model

- **Model type:** Random Forest Classifier (with median/mean imputation for missing values)
- **Target:** TenYearCHD — risk of coronary heart disease within 10 years
- **Training approach:** minority class (CHD = Yes) was oversampled with `RandomOverSampler` before training; hyperparameters were tuned via 5-fold `GridSearchCV` optimizing ROC-AUC.
- **Best hyperparameters:** {'randomforestclassifier__max_depth': 30, 'randomforestclassifier__n_estimators': 100, 'simpleimputer__strategy': 'median'}
- **Held-out test accuracy:** 83.7%